

ADVANCED

Stage 6

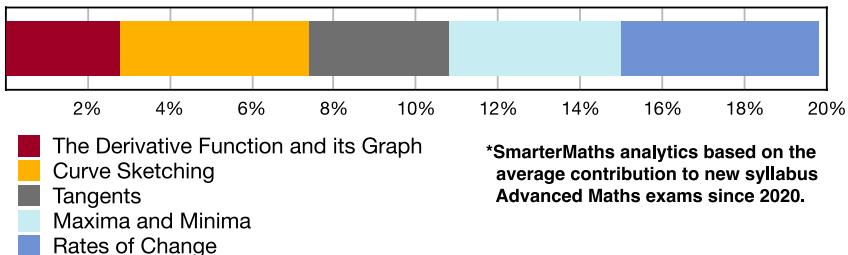
Calculus (Adv), C3 Applications of Calculus (Adv)

Tangents (Y12)

Teacher: SHS

Exam Equivalent Time: 72 minutes (based on allocation of 1.5 minutes per mark)

C3 Applications of Calculus



HISTORICAL CONTRIBUTION

- C3 Applications of Calculus* is the 800 pound gorilla in the Advanced rainforest, contributing a massive 19.8% to new syllabus exams since introduced in 2020.
- This topic has been split into five sub-topics for analysis purposes: 1-*The Derivative Function and its Graph* (2.8%), 2-*Curve Sketching* (4.6%), 3-*Tangents* (3.4%), 4-*Maxima and Minima* (4.2%) and 5-*Rates of Change* (4.8%).
- This analysis looks at *Tangents*.

HSC ANALYSIS - What to expect and common pitfalls

- Tangents* (3.4%) appear in two key areas of the syllabus: *C1 Introduction to Calculus* (involving standard differentiation - 0.8%) and *C3 Applications of Calculus* (involving log, exponential and trig underlyings - 3.4%).
- This analysis looks at *Tangents* as a whole, which have been examined in 4 out of the 5 new syllabus exams to date. Worth noting the 2020 and 2021 exams each had multiple questions on this sub-topic which were poorly answered and represent key revision.
- 2024 Adv 29 is a must review question which tested this sub-topic in a novel way, requiring a good understanding of both tangents and normals.
- Composite functions overlapped with tangents in 2021 9 MC - differentiating these types of functions is an important competency in the new syllabus and this needs to be carefully reviewed.

Questions

1. Calculus, 2ADV C3 SM-Bank 1 MC

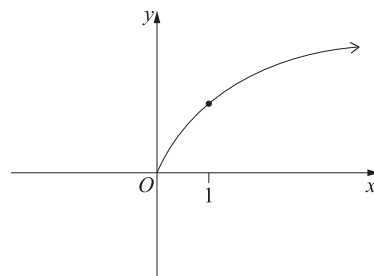
The tangent to the graph of $y = x^3 - ax^2 + 1$ at $x = 1$ passes through the origin.

The value of a is

- $\frac{1}{2}$
- 1
- $\frac{3}{2}$
- $\frac{5}{2}$

2. Calculus, 2ADV C3 2020 HSC 8 MC

The graph of $y = f(x)$ is shown.



Which of the following inequalities is correct?

- $f''(1) < 0 < f'(1) < f(1)$
- $f''(1) < 0 < f(1) < f'(1)$
- $0 < f''(1) < f'(1) < f(1)$
- $0 < f''(1) < f(1) < f'(1)$

3. Calculus, 2ADV C3 2021 HSC 9 MC

Let $h(x) = f(g(x))$ where the function $f(x)$ is an odd function and the function $g(x)$ is an even function.

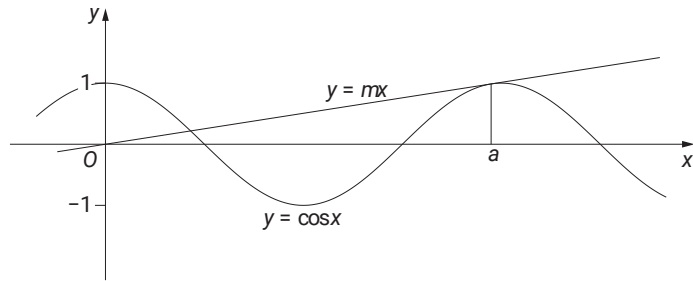
The tangent to $y = h(x)$ at $x = k$, where $k > 0$, has the equation $y = mx + c$.

What is the equation of the tangent to $y = h(x)$ at $x = -k$?

- $y = mx + c$
- $y = -mx + c$
- $y = mx - c$
- $y = -mx - c$

4. Calculus, 2ADV C3 2021 HSC 10 MC

The line $y = mx$ is a tangent to the curve $y = \cos x$ at the point where $x = a$, as shown in the diagram.



Which of the following statements is true?

- A. $m < \frac{1}{a} < \frac{1}{2\pi}$
- B. $\frac{1}{2\pi} < m < \frac{1}{a}$
- C. $\frac{1}{2\pi} < \frac{1}{a} < m$
- D. $m < \frac{1}{2\pi} < \frac{1}{a}$

5. Calculus, 2ADV C3 2005 HSC 2d

Find the equation of the tangent to $y = \log_e x$ at the point $(e, 1)$. (2 marks)

6. Calculus, 2ADV C3 2006 HSC 2c

Find the equation of the tangent to the curve $y = \cos 2x$ at the point whose x -coordinate is $\frac{\pi}{6}$. (3 marks)

7. Calculus, 2ADV C3 2007 HSC 2c

The point $P(\pi, 0)$ lies on the curve $y = x \sin x$. Find the equation of the tangent to the curve at P . (3 marks)

8. Calculus, 2ADV C3 2021 HSC 13

Find the exact gradient of the tangent to the curve $y = x \tan x$ at the point where $x = \frac{\pi}{3}$. (3 marks)

9. Calculus, 2ADV C3 2016 HSC 11f

Find the gradient of the tangent to the curve $y = \tan x$ at the point where $x = \frac{\pi}{8}$.

Give your answer correct to 3 significant figures. (2 marks)

10. Calculus, 2ADV C3 2005 HSC 5c

Find the coordinates of the point P on the curve $y = 2e^x + 3x$ at which the tangent to the curve is parallel to the line $y = 5x - 3$. (3 marks)

11. Calculus, 2ADV C3 2018 HSC 12b

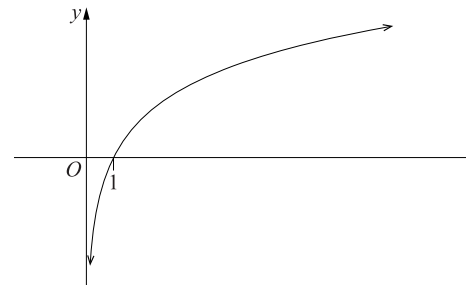
Find the equation of the tangent to the curve $y = \cos 2x$ at $x = \frac{\pi}{6}$. (3 marks)

12. Calculus, 2ADV C3 2010 HSC 2c

Find the gradient of the tangent to the curve $y = \ln(3x)$ at the point where $x = 2$. (2 marks)

13. Calculus, 2ADV C3 2020 HSC 29

The diagram shows the graph of $y = c \ln x$, $c > 0$.



a. Show that the equation of the tangent to $y = c \ln x$ at $x = p$, where $p > 0$, is

$$y = \frac{c}{p}x - c + c \ln p. \quad (2 \text{ marks})$$

b. Find the value of c such that the tangent from part (a) has a gradient of 1 and passes through the origin. (2 marks)

14. Calculus, 2ADV C3 2014 HSC 15c

The line $y = mx$ is a tangent to the curve $y = e^{2x}$ at a point P .

- i. Sketch the line and the curve on one diagram. (1 mark)
- ii. Find the coordinates of P . (3 marks)
- iii. Find the value of m . (1 mark)

15. Calculus, 2ADV C3 2021 HSC 31

By considering the equation of the tangent to $y = x^2 - 1$ at the point $(a, a^2 - 1)$, find the equations of the two tangents to $y = x^2 - 1$ which pass through $(3, -8)$. (4 marks)

16. Calculus, 2ADV C3 2024 HSC 29

Consider the curve $y = ax^2 + bx + c$, where $a \neq 0$.

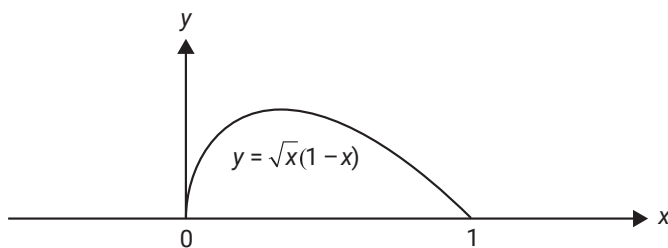
At a particular point, the tangent and normal to the curve are given by $t(x) = 2x + 3$ and $n(x) = -\frac{1}{2}x - 2$ respectively.

The curve has a minimum turning point at $x = -4$.

Find the values of a , b and c . (4 marks)

17. Calculus, 2ADV C3 SM-Bank 7

The graph of $f(x) = \sqrt{x}(1-x)$ for $0 \leq x \leq 1$ is shown below.

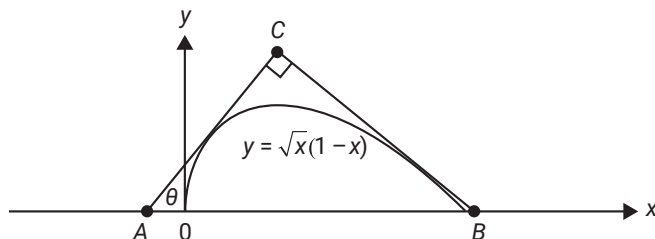


i. Calculate the area between the graph of $f(x)$ and the x -axis. (2 marks)

ii. For x in the interval $(0, 1)$, show that the gradient of the tangent to the graph of $f(x)$ is $\frac{1-3x}{2\sqrt{x}}$. (1 mark)

The edges of the right-angled triangle ABC are the line segments AC and BC , which are tangent to the graph of $f(x)$, and the line segment AB , which is part of the horizontal axis, as shown below.

Let θ be the angle that AC makes with the positive direction of the horizontal axis.



iii. Find the equation of the line through B and C in the form $y = mx + c$, for $\theta = 45^\circ$. (3 marks)

Worked Solutions

1. Calculus, 2ADV C3 SM-Bank 1 MC

$$y = x^3 - ax^2 + 1$$

$$\frac{dy}{dx} = 3x^2 - 2ax$$

$$\text{At } x = 1 \Rightarrow y = 2 - a, \quad \frac{dy}{dx} = 3 - 2a$$

Tangent passes through $(1, 2 - a)$ and $(0, 0)$

$$m_{\text{tang}} = 2 - a$$

Equating gradients:

$$3 - 2a = 2 - a$$

$$\therefore a = 1$$

$$\Rightarrow B$$

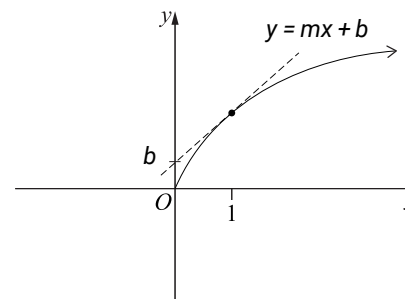
2. Calculus, 2ADV C3 2020 HSC 8 MC

$y = f(x)$ is concave down at $x = 1$

$$f''(1) < 0$$

$\Rightarrow$ Eliminate C and D.

Mean mark 52%.



$$\text{Tangent } \Rightarrow y = mx + b$$

$$\text{At } x = 1, m_{\text{tang}} = f'(1)$$

At intersection:

$$mx + b = f(x)$$

$$f'(1) \times 1 + b = f(1)$$

$$\therefore f'(1) < f(1), \quad (b > 0)$$

$$\therefore f''(1) < 0 < f'(1) < f(1)$$

$$\Rightarrow A$$

COMMENT: Note the concavity causes any tangent to cut the positive y -axis. No y -axis scale is therefore required.

3. Calculus, 2ADV C3 2021 HSC 9 MC

$h(x) = f(g(x))$

$h'(x) = g'(x) \cdot f'(g(x))$

At $x = k,$

$h(k) = f(g(k))$

$h'(k) = g'(x) \cdot f'(g(x)) = m$

At $x = -k,$

$h(-k) = f(g(-k)) = f(g(k)) = h(k)$

$\rightarrow c$ is the same

$h'(-k) = g'(-k) \cdot f'(g(-k))$

$= -g'(k) \cdot f'(g(k)) \quad (g(k) \text{ is even} \rightarrow g'(-k) = -g'(k))$

$= -m$

$\therefore$ Equation of tangent is $y = -mx + c$

$\Rightarrow B$

Mean mark 52%.

4. Calculus, 2ADV C3 2021 HSC 10 MC

By Elimination:

$a < 2\pi \Rightarrow \frac{1}{a} > \frac{1}{2\pi}$

$\rightarrow$ Eliminate A

$m \approx \frac{1}{2\pi} \approx \frac{1}{6}$

$\frac{1}{a}$ is much closer to 1 (by inspection)

$\frac{1}{a} > m$

$\rightarrow$ Eliminate C

At $x = 2\pi,$

$y = \cos 2\pi = 1$ and $y = mx > 1$

$m > \frac{1}{2\pi}$

$\therefore \frac{1}{2\pi} < m < \frac{1}{a}$

$\Rightarrow B$

◆◆◆Mean mark 27%.

5. Calculus, 2ADV C3 2005 HSC 2d

$y = \log_e x$

$\frac{dy}{dx} = \frac{1}{x}$

At $(e, 1),$

$m = \frac{1}{e}$

Equation of tangent, $m = \frac{1}{e},$ through $(e, 1)$

$y - y_1 = m(x - x_1)$

$y - 1 = \frac{1}{e}(x - e)$

$y - 1 = \frac{x}{e} - 1$

$y = \frac{x}{e}$

6. Calculus, 2ADV C3 2006 HSC 2c

$$y = \cos 2x$$

$$\frac{dy}{dx} = -2\sin 2x$$

$$\text{When } x = \frac{\pi}{6}$$

$$y = \cos\left(2 \times \frac{\pi}{6}\right)$$

$$= \cos\left(\frac{\pi}{3}\right)$$

$$= \frac{1}{2}$$

$$\frac{dy}{dx} = -2\sin\left(\frac{\pi}{3}\right)$$

$$= -2 \times \frac{\sqrt{3}}{2}$$

$$= -\sqrt{3}$$

Equation of tangent, $m = -\sqrt{3}$, through $\left(\frac{\pi}{6}, \frac{1}{2}\right)$:

$$y - y_1 = m(x - x_1)$$

$$y - \frac{1}{2} = -\sqrt{3}\left(x - \frac{\pi}{6}\right)$$

$$y - \frac{1}{2} = -\sqrt{3}x + \frac{\sqrt{3}\pi}{6}$$

$$y = -\sqrt{3}x + \left(\frac{\sqrt{3}\pi}{6} + \frac{1}{2}\right)$$

7. Calculus, 2ADV C3 2007 HSC 2c

$$y = x \sin x$$

$$\frac{dy}{dx} = x \times \frac{d}{dx}(\sin x) + \frac{d}{dx}x \times \sin x$$

$$= x \cos x + \sin x$$

$$\text{When } x = \pi$$

$$\frac{dy}{dx} = \pi \times \cos \pi + \sin \pi$$

$$= \pi(-1) + 0$$

$$= -\pi$$

Equation of line, $m = -\pi$, through $P(\pi, 0)$:

$$y - y_1 = m(x - x_1)$$

$$y - 0 = -\pi(x - \pi)$$

$$\therefore y = -\pi x + \pi^2$$

8. Calculus, 2ADV C3 2021 HSC 13

$$y = x \tan x$$

$$\frac{dy}{dx} = \tan x + x \sec^2 x$$

Find m when $x = \frac{\pi}{3}$:

$$\frac{dy}{dx} = \tan \frac{\pi}{3} + \frac{\pi}{3} \cdot \frac{1}{\cos^2 \frac{\pi}{3}}$$

$$= \sqrt{3} + \frac{\pi}{3} \cdot \frac{1}{\frac{1}{4}}$$

$$= \sqrt{3} + \frac{4\pi}{3}$$

9. Calculus, 2ADV C3 2016 HSC 11f

$y = \tan x = \frac{\sin x}{\cos x}$

$\frac{dy}{dx} = \sec^2 x$

When $x = \frac{\pi}{8}$,

$$\begin{aligned} \therefore \frac{dy}{dx} &= \sec^2\left(\frac{\pi}{8}\right) \\ &= \frac{1}{\cos^2\left(\frac{\pi}{8}\right)} \\ &= 1.1715\dots \\ &= 1.17 \text{ (to 3 sig fig)} \end{aligned}$$

10. Calculus, 2ADV C3 2005 HSC 5c

Gradient of $y = 5x - 3$ is 5.

$y = 2e^x + 3x$

$\frac{dy}{dx} = 2e^x + 3$

Find x when $\frac{dy}{dx} = 5$

$$\begin{aligned} 5 &= 2e^x + 3 \\ 2e^x &= 2 \\ e^x &= 1 \\ x &= 0 \end{aligned}$$

When $x = 0$

$$\begin{aligned} y &= 2e^0 + (3 \times 0) \\ &= 2 \end{aligned}$$

$\therefore P$ has coordinates $(0, 2)$

11. Calculus, 2ADV C3 2018 HSC 12b

$y = \cos 2x$

$\frac{dy}{dx} = -2\sin 2x$

When $x = \frac{\pi}{6}$:

$y = \cos \frac{\pi}{3} = \frac{1}{2}$

$\frac{dy}{dx} = -\sin \frac{\pi}{3} = -\sqrt{3}$

Equation of tangent $m = -\sqrt{3}$, through $\left(\frac{\pi}{6}, \frac{1}{2}\right)$:

$y - y_1 = m(x - x_1)$

$y - \frac{1}{2} = -\sqrt{3}\left(x - \frac{\pi}{6}\right)$

$y = -\sqrt{3}x + \frac{\sqrt{3}\pi}{6} + \frac{1}{2}$

$\therefore y = -\sqrt{3}x + \frac{\sqrt{3}\pi + 3}{6}$

12. Calculus, 2ADV C3 2010 HSC 2c

$y = \ln(3x)$

$\frac{dy}{dx} = \frac{3}{3x} = \frac{1}{x}$

At $x = 2$,

$\frac{dy}{dx} = \frac{1}{2}$

$\therefore$ The gradient at $x = 2$ is $\frac{1}{2}$.

CAUTION: Read the question carefully! MANY wasted valuable exam time finding the equation of the tangent here.

13. Calculus, 2ADV C3 2020 HSC 29

a. $y = c\ln x$

$$\frac{dy}{dx} = \frac{c}{x}$$

At $x = p$,

$$m_{\text{tang}} = \frac{c}{p}$$

Tangent passes through $(p, c\ln p)$

$\therefore$ Equation of tangent

$$y - c\ln p = \frac{c}{p}(x - p)$$

$$y = \frac{c}{p}x - c + c\ln p$$

b. If $m_{\text{tang}} = 1$,

$$\frac{c}{p} = 1$$

$$c = p$$

If tangent passes through $(0, 0)$

$$0 = -c + c\ln c$$

$$0 = c(\ln c - 1)$$

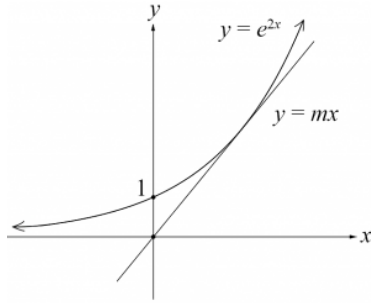
$$\ln c = 1 \quad (c > 0)$$

$$\therefore c = e$$

♦ Mean mark part (b) 40%.

14. Calculus, 2ADV C3 2014 HSC 15c

i.



ii.

$$y = e^{2x}$$

$$\frac{dy}{dx} = 2e^{2x}$$

Gradient of $y = mx$ is m

Gradients equal when

$$2e^{2x} = m$$

$$e^{2x} = \frac{m}{2}$$

$$\ln e^{2x} = \ln\left(\frac{m}{2}\right)$$

$$2x = \ln\left(\frac{m}{2}\right)$$

$$x = \frac{1}{2}\ln\left(\frac{m}{2}\right)$$

$$\text{When } x = \frac{1}{2}\ln\left(\frac{m}{2}\right)$$

$$y = e^{2 \times \frac{1}{2}\ln\left(\frac{m}{2}\right)}$$

$$= e^{\ln\left(\frac{m}{2}\right)}$$

$$= \frac{m}{2}$$

$$\therefore P\left(\frac{1}{2}\ln\left(\frac{m}{2}\right), \frac{m}{2}\right)$$

iii. $y = mx$ passes through $(0, 0)$ and $\left(\frac{1}{2}\ln\left(\frac{m}{2}\right), \frac{m}{2}\right)$

Equating gradients:

$$\frac{\frac{m}{2} - 0}{\frac{1}{2}\ln\left(\frac{m}{2}\right) - 0} = m$$

$$\frac{m}{2} = m \times \frac{1}{2}\ln\left(\frac{m}{2}\right)$$

♦ Mean mark 40%

COMMENT: Given $y = e^{\ln\left(\frac{m}{2}\right)}$, it follows $y = \frac{m}{2}$. Make sure you understand the arithmetic behind this (NB. Simply take the $\ln$ of both sides).

♦♦ Mean mark 30%.

$$\ln\left(\frac{m}{2}\right) = 1$$

$$\frac{m}{2} = e^1$$

$$m = 2e$$

15. Calculus, 2ADV C3 2021 HSC 31

$$y = x^2 - 1$$

$$\frac{dy}{dx} = 2x$$

$$\text{At } x = a, \frac{dy}{dx} = 2a$$

♦♦ Mean mark 36%.

Find equation of line $m = 2a$, through $(a, a^2 - 1)$:

$$y - (a^2 - 1) = 2a(x - a)$$

$$y - a^2 + 1 = 2ax - 2a^2$$

$$y = 2ax - a^2 - 1$$

If tangent passes through $(3, -8)$:

$$2a(3) - a^2 - 1 = -8$$

$$6a - a^2 + 7 = 0$$

$$a^2 - 6a - 7 = 0$$

$$(a - 7)(a + 1) = 0$$

$$\Rightarrow a = 7 \text{ or } -1$$

$\therefore$ Equation of tangents:

$$y = 14x - 50$$

$$y = -2x - 2$$

16. Calculus, 2ADV C3 2024 HSC 29

$$y = ax^2 + bx + c$$

$$y' = 2ax + b$$

SP's when $y' = 0$ and $x = -4$:

$$-8a + b = 0 \dots (1)$$

Find point where $t(x), n(x)$ and curve intersect:

$$2x + 3 = -\frac{1}{2}x - 2$$

$$4x + 6 = -x - 4$$

$$5x = -10$$

$$x = -2$$

$$\Rightarrow \text{Intersection at } (-2, -1)$$

$$\text{At } x = -2, m_{\text{tang}} = -4a + b$$

$$-4a + b = 2 \dots (2)$$

Subtract (2) - (1):

$$4a = 2 \Rightarrow a = \frac{1}{2}$$

Substitute $a = \frac{1}{2}$ into (1):

$$\Rightarrow b = 4$$

Since $(-2, -1)$ lies on y :

$$-1 = \frac{1}{2}(-2)^2 + 4(-2) + c$$

$$\Rightarrow c = 5$$

$$\therefore a = \frac{1}{2}, b = 4, c = 5$$

17. Calculus, 2ADV C3 SM-Bank 7

$$\begin{aligned} \text{i. Area} &= \int_0^1 (\sqrt{x} - x\sqrt{x}) \, dx \\ &= \int_0^1 \left(x^{\frac{1}{2}} - x^{\frac{3}{2}}\right) \, dx \\ &= \left[\frac{2}{3}x^{\frac{3}{2}} - \frac{2}{5}x^{\frac{5}{2}}\right]_0^1 \\ &= \left(\frac{2}{3} - \frac{2}{5}\right) - (0 - 0) \\ &= \frac{10}{15} - \frac{6}{15} \\ &= \frac{4}{15} \text{ units}^2 \end{aligned}$$

$$\begin{aligned} \text{ii. } f(x) &= x^{\frac{1}{2}} - x^{\frac{3}{2}} \\ f'(x) &= \frac{1}{2}x^{-\frac{1}{2}} - \frac{3}{2}x^{\frac{1}{2}} \\ &= \frac{1}{2\sqrt{x}} - \frac{3\sqrt{x}}{2} \\ &= \frac{1-3x}{2\sqrt{x}} \dots \text{as required.} \end{aligned}$$

$$\begin{aligned} \text{iii. } m_{AC} &= \tan 45^\circ = 1 \\ \Rightarrow m_{BC} &= -1 \quad (m_{BC} \perp m_{AC}) \end{aligned}$$

At point of tangency of BC , $f'(x) = -1$

$$\begin{aligned} \frac{1-3x}{2\sqrt{x}} &= -1 \\ 1-3x &= -2\sqrt{x} \\ 3x-2\sqrt{x}-1 &= 0 \end{aligned}$$

Let $a = \sqrt{x}$,

$$\begin{aligned} 3a^2 - 2a - 1 &= 0 \\ (3a+1)(a-1) &= 0 \\ a &= 1 \text{ or } -\frac{1}{3} \end{aligned}$$

$$\begin{aligned} \therefore \sqrt{x} &= 1 \quad \text{or} \quad \sqrt{x} = -\frac{1}{3} \quad (\text{no solution}) \\ x &= 1 \end{aligned}$$

$$f(1) = \sqrt{1}(1-1) = 0 \Rightarrow B(1, 0)$$

◆◆◆ Mean mark (Vic) part (iii) 20%.
MARKER'S COMMENT: Most successful answers introduced a pronumeral such as $a = \sqrt{x}$ to solve.

Equation of BC , $m = -1$, through $(1,0)$:

$$y - 0 = -1(x - 1)$$

$$y = -x + 1$$

